Title of Topic: Keys and Integrity Constraints

Course Code: 23IT201

Course Title: DataBase Management Systems

Session Number: 6

Academic Year: 2024 - 2025 (Even Sem)

Module 1 – INTRODUCTION

1. Introduction to DBMS, Characteristics of DBMS, DBMS vs File Systems, need for DBMS.
2. Three Level DBMS Architecture.
3. **Data Models, Introduction, Benefits, and Phases. ER Diagrams – Symbols, Components, Relationships, Weak entities, Attributes, Cardinality.**

4.Relational Algebra, Domain Relational Calculus, Tuple Relational Calculus,

5. Keys - primary Key, Foreign Key.

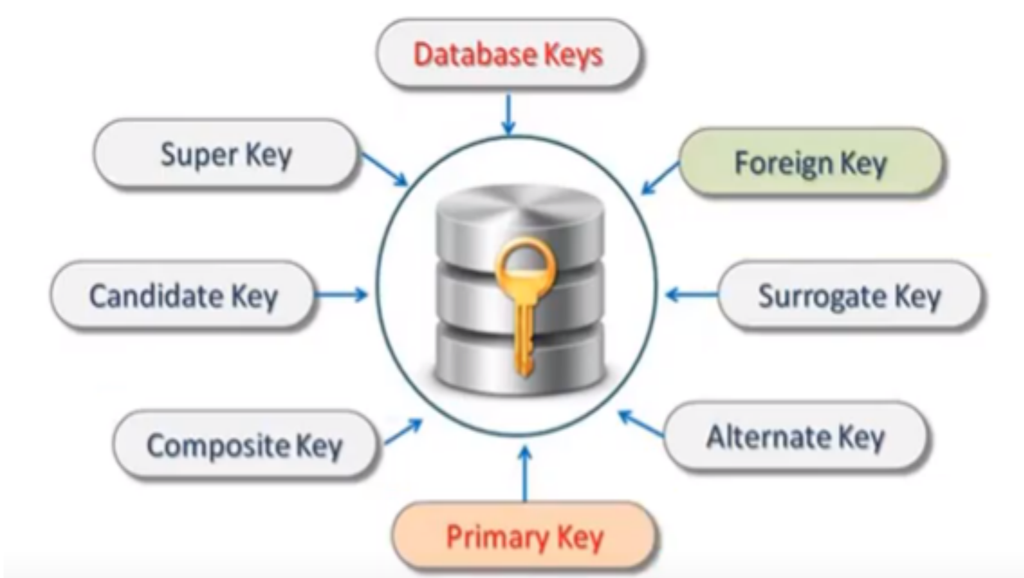
Keys and Integrity Constraints

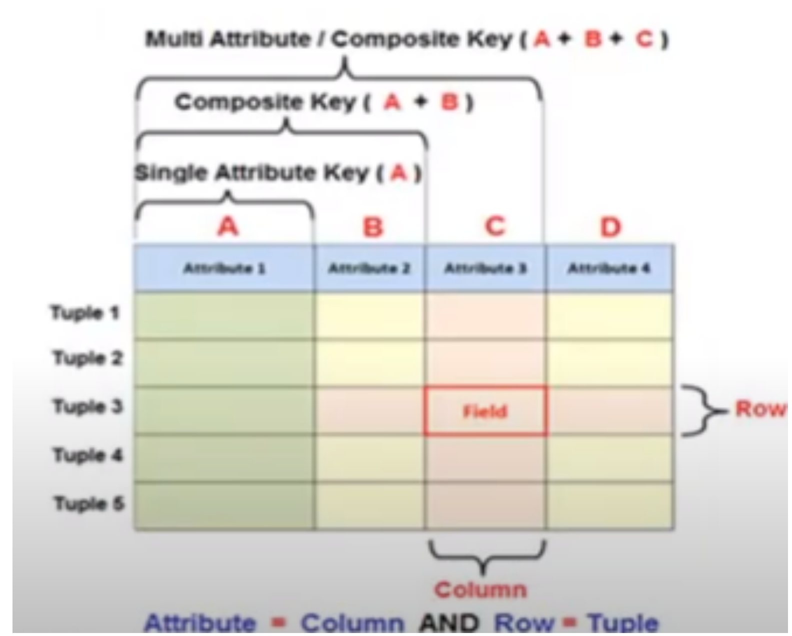
Topics to be Covered

- Keys in DBMS
- Integrity Constraints
- Domain Constraints
- Entity Integrity Constraints
- Referential Integrity Constraints
- Key Constraints

Keys in DBMS

- Ensure that a table record can be uniquely identified
- Allows to establish a relationship between tables and identify the relation between tables
- Help to enforce identity and integrity in the relationship





Keys in DBMS:

Keys in DBMS

Super Key

- Every key in a table
- An attribute or a set of attributes that can be used to identify rows in a table
- May have additional attributes that are not needed for unique identification

Student Table

SID	REG_ID	NAME	BRANCH	EMAIL
1	CS-2019-37	John	CS	john@xyz.com
2	CS-2018-02	Adam	CS	adamcool@xyz.com
3	IT-2019-01	Adam	IT	adamnerd@xyz.com
4	ECE-2019-07	Elly	ECE	elly@xyz.com

Super Keys

- SID
- REG_ID
- EMAIL
- SID + REG_ID
- REG_ID + EMAIL
- EMAIL + SID
- SID + REG_ID + EMAIL

Keys in DBMS

Candidate Key

- Set of attributes that uniquely identify tuples in a table
- A super key with no repeated attributes
- Minimal subset of super key
- If any proper subset of a super key is a super key then that key cannot be a candidate key

Student Table

SID	REG_ID	NAME	BRANCH	EMAIL
1	CS-2019-37	John	CS	john@xyz.com
2	CS-2018-02	Adam	CS	adamcool@xyz.com
3	IT-2019-01	Adam	IT	adamnerd@xyz.com
4	ECE-2019-07	Elly	ECE	elly@xyz.com

Candidate Keys

- SID
- REG_ID
- EMAIL

We can choose any one as Candidate Key

Keys in DBMS

Properties of Candidate Key

- It must contain unique values
- Candidate key may have multiple attributes
- Must not contain null values
- It should contain minimum fields to ensure uniqueness
- Uniquely identify each record in a table

Keys in DBMS

Primary Key

- An attribute or a set of attributes in a table that uniquely identify every row in that table
- Can't be a duplicate meaning the same value can't appear more than once in the table
- A table cannot have more than one primary key

Primary Key

Student Table

SID	REG_ID	NAME	BRANCH	EMAIL
1	CS-2019-37	John	CS	john@xyz.com
2	CS-2018-02	Adam	CS	adamcool@xyz.com
3	IT-2019-01	Adam	IT	adamnerd@xyz.com
4	ECE-2019-07	Elly	ECE	elly@xyz.com

Candidate Keys:

- SID
- REG_ID
- EMAIL

Pick any one as Primary Key

Keys in DBMS

Rules for defining Primary Key

- Two rows can't have the same primary key value
- It is must for every row to have a primary key value
- The primary key field cannot be null
- The value in a primary key column can never be modified or updated if any foreign key refers to that primary key

Keys in DBMS

Alternate Key

- An attribute or a set of attributes in a table that uniquely identify every row in that table
- A table can have multiple choices for a primary key but only one can be set as the primary key
- All the keys which are not primary key are called an Alternate Key

Alternate Key

Student Table

SID	REG_ID	NAME	BRANCH	EMAIL
1	CS-2019-37	John	CS	john@xyz.com
2	CS-2018-02	Adam	CS	adamcool@xyz.com
3	IT-2019-01	Adam	IT	adamnerd@xyz.com
4	ECE-2019-07	Elly	ECE	elly@xyz.com

- SID
- REG_ID
- EMAIL

If we choose REG_ID as Primary Key then SID and EMAIL will become Alternate Key

Keys in DBMS

Foreign Key

- An attribute that creates a relationship between two tables
- Maintain data integrity and allow navigation between two different instances of an entity
- Acts as a cross-reference between two tables as it references the primary key of another table

Foreign key → BRANCH_CODE

Student Table

SID	REG_ID	NAME	BRANCH_CODE	EMAIL
1	CS-2019-37	John	CS	john@xyz.com
2	CS-2018-02	Adam	CS	adamcool@xyz.com
3	IT-2019-01	Adam	IT	adamnerd@xyz.com
4	ECE-2019-07	Elly	ECE	elly@xyz.com

DB Error

Branch Table

Update an Entry which is referred in Student Table

BRANCH_CODE	BRANCH_NAME	HOD	...
CS	Computer Science	John Mayer	...
ME	Mechanical Engineering	Adam Lee	...
IT	Information Technology	L. Subramanyam	...
ECE	Electronics and Communication	Lilibet Windsor	...

Keys in DBMS

Composite Key

- Combination of two or more attributes that uniquely identify rows in a table
- The combination of attributes guarantees uniqueness, though individually uniqueness is not guaranteed
- Hence, they are combined to uniquely identify records in a table

E.g., (SID,REG_ID),(REG_ID,EMAIL),(EMAIL,SID),(SID,REG_ID,EMAIL) etc… all are composite keys

Student Table

SID	REG_ID	NAME	BRANCH	EMAIL
1	CS-2019-37	John	CS	john@xyz.com
2	CS-2018-02	Adam	CS	adamcool@xyz.com
3	IT-2019-01	Adam	IT	adamnerd@xyz.com
4	ECE-2019-07	Elly	ECE	elly@xyz.com

Keys in DBMS

Compound Key

- Two or more attributes that allows to uniquely recognize a specific record
- Each attribute may not be unique by itself within the database
- However, when combined with the other attribute or attributes the combination of composite keys become unique

E.g., (REG_ID,BRANCH_CODE) is a compound key because branch attribute is a foreign key

Student Table

SID	REG_ID	NAME	BRANCH	EMAIL
1	CS-2019-37	John	CS	john@xyz.com
2	CS-2018-02	Adam	CS	adamcool@xyz.com
3	IT-2019-01	Adam	IT	adamnerd@xyz.com
4	ECE-2019-07	Elly	ECE	elly@xyz.com

Branch Table

BRANCH_CODE	BRANCH_NAME	HOD	...
CS	Computer Science	John Mayer	...
ME	Mechanical Engineering	Adam Lee	...
IT	Information Technology	L. Subramanyam	...
ECE	Electronics and Communication	Lilibet Windsor	...

Keys in DBMS

Surrogate Key

- If a relation has no attribute which can be used to identify the data stored in it, then we create an attribute for this purpose
- Adds no meaning to the data but serves the sole purpose of identifying rows uniquely in a table
- Created when there is no natural primary key
- Do not lend any meaning to the data in the table and is usually an integer
- Value generated right before the record is inserted into a table

Keys in DBMS

Surrogate Key

- Allowed when

✓ No property has the parameter of the primary key

✓ In the table when the primary key is too big or complicated

Student Count	Student ID	First Name	SSN	Course
1	CS 1001	S1	CA245	Java
2	EE 1002	S2	WA465	Python
3	EE 1003	S3	AR365	Python
4	CS 1002	S4	WA764	Java
5	IE 1005	S5	CA453	C++
6	CS 1003	S6	CA924	Java

Surrogate Key Column Added As Primary Key

Integrity Constraints

- Set of rules

✓ Used to maintain the quality of information

✓ DataBase is not permitted to violate

- Ensure that inserting data, updating data and other processes have to be performed in such a way that data integrity is not affected
- Used to guard against accidental damage to the database
- Used to ensure accuracy and consistency of the data in a relational database
- Every relation has some constraints (condition) to make a relation valid is known as relational integrity constraints

Integrity Constraints:

Integrity Constraints

Types of Integrity Constraints

- Domain Constraints
- Entity Integrity Constraints
- Referential Integrity Constraints
- Key Constraints

Domain Constraints

- Definition of a valid set of values for an attribute
- Defines data type, length or size
- Data type of domain constraints include string, character, integer, time, date, currency, etc…
- Value of the attribute must be available in the corresponding domain

E.g., If there is a column (attribute) of integer values then alphabet cannot be placed in it, since the attribute had only integer values as its domain

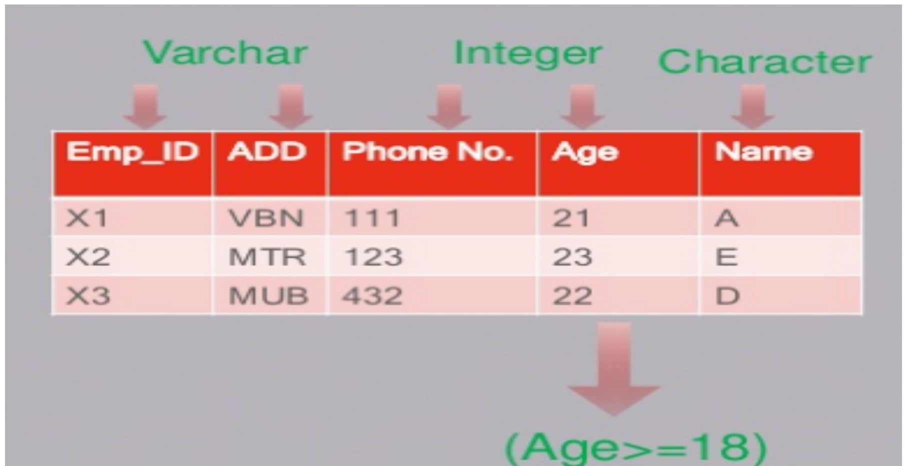
STUDENT_ID	NAME	SEMESTER	AGE
101	Manish	1st	18
102	Rohit	3rd	19
103	Badal	5th	20
104	Amit	7th	A

Not allowed. Because AGE is an integer value

Domain Constraints

- Every attribute is bound to have specific range and/or specific values

E.g., Age cannot be 0 or less, Name cannot be in integer format



Emp_ID	ADD	Phone No.	Age	Name
X1	VBN	111	21	A
X2	MTR	123	23	E
X3	MUB	432	22	D

- Null value is allowed
- The value is unique or not for an attribute

Domain Constraints

E.g., CREATE TABLE persons(ID int NOT NULL, Name varchar(255) NOT NULL, Age int, CHECK (Age>=18));

Output

```
Table created.
```

DESC persons;

TABLE PERSONS

Column	Null?	Type
ID	NOT NULL	NUMBER
NAME	NOT NULL	VARCHAR2(255)
AGE	-	NUMBER

Download CSV

3 rows selected.

INSERT INTO persons VALUES('1','','26');

```
ORA-01400: cannot insert NULL into ("SQL_LHHGHTQALMCMCEVKLWKDDKTZZ"."PERSONS"."NAME") ORA-06512: at "SYS.DBMS_SQL", line 1721
```

Domain Constraints

INSERT INTO persons VALUES('1','Sindhu','26');

```
1 row(s) inserted.
```

INSERT INTO persons VALUES('2','Karthi','18');

```
1 row(s) inserted.
```

INSERT INTO persons VALUES('3','Sandhya','15');

```
ORA-02290: check constraint (SQL_LHHGHTQALMCMCEVKLWKDDKTZZ.SYS_C0048236534) violated ORA-06512: at "SYS.DBMS_SQL", line 1721
```

SELECT * FROM persons;

ID	NAME	AGE
1	Sindhu	26
2	Karthi	18

Download CSV

2 rows selected.

Entity Integrity Constraints

- States that primary key value cannot be null
- Because the primary key value is used to identify individual rows in relation and if the primary key has a null value, then those rows cannot be identified
- Table can contain a null value other than the primary key field
- Null value is different from zero value or space
- Entity integrity constraints assure that a specific row in a table can be identified

Null value not allowed

REG.NO	MODEL_id	RATE
ABC-112	1	45,00$
ACC-223	6	65,00$
BAA-441	2	57,00$
NULL	5	35,00$
CCE-325	3	45,00$
CCE-326	4	61,00$

EMP_ID	EMP_NAME	SALARY
111	Mohan	20000
112	Rohan	30000
113	Sohan	35000
	Logan	20000

Not allowed as Primary Key can't contain NULL value

Entity Integrity Constraints

E.g., CREATE TABLE persons(ID int PRIMARY KEY, LastName varchar(255) NOT NULL, FirstName varchar(255),Age int);

Output

```
Table created.
```

INSERT INTO persons VALUES('1','ranjith','sindhu','26');

```
1 row(s) inserted.
```

INSERT INTO persons VALUES('1','karthi','sindhu','28');

```
ORA-00001: unique constraint (SQL_XWUKQOOFTQNWPZUHEGBPQRGKY.SYS_C0048214477) violated ORA-06512: at "SYS.DBMS_SQL", line 1721
```

Entity Integrity Constraints

E.g., CREATE TABLE persons(ID int NOT NULL, LastName varchar(255) NOT NULL, FirstName varchar(255),Age int,CONSTRAINT PK_person PRIMARY KEY(ID,LastName));

Output

```
Table created.
```

INSERT INTO persons VALUES('1','ranjith','sindhu','26');

```
1 row(s) inserted.
```

INSERT INTO persons VALUES(‘2','ranjith','karthi','30');

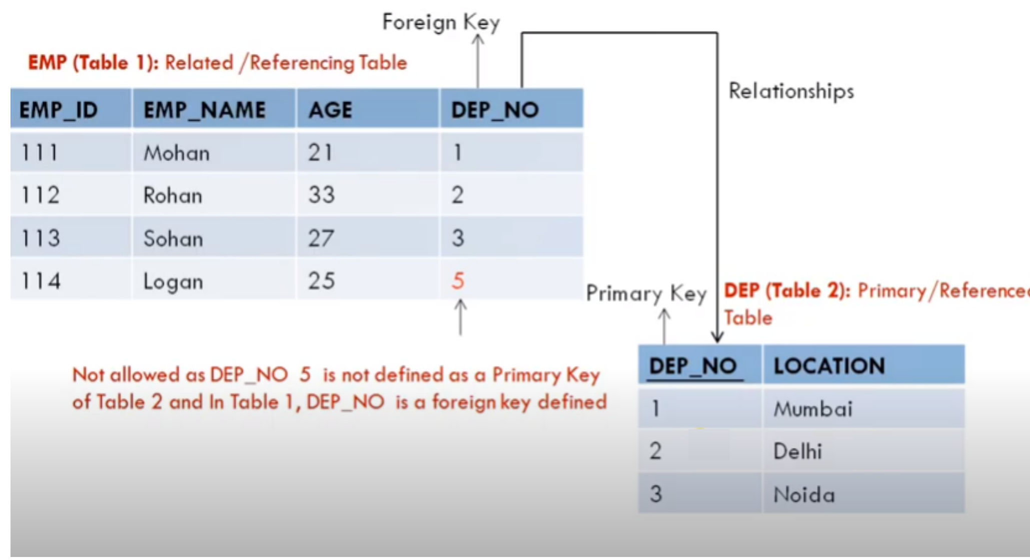
```
ORA-00001: unique constraint (SQL_BMKCJDUGBOCTWCVNRLFAUIIQT.PK_PERSON) violated ORA-06512: at "SYS.DBMS_SQL", line 1721
```

Referential Integrity Constraints

- Specified between two tables and is used to maintain the consistency among rows between the two tables
- Implemented through foreign key, which is a key attribute of a relation that can be referred in other relation
- Foreign key is a field (or collection of fields) in one table that refers to the primary key in another table

REG.NO	MODEL_id	RATE
ABC-112	1	45,00$
ACC-223	6	65,00$
BAA-441	2	57,00$
BSA-224	5	35,00$
CCE-325	3	45,00$
CCE-326	4	61,00$

Model_id	Mark	Model
1.	ford	focus
2.	ford	mondeo
3.	Peugeot	307
4.	Peugeot	407



EMP_ID	EMP_NAME	AGE	DEP_NO
111	Mohan	21	1
112	Rohan	33	2
113	Sohan	27	3
114	Logan	25	5

DEP_NO	LOCATION
1	Mumbai
2	Delhi
3	Noida

Referential Integrity Constraints

Rules

- A record cannot be deleted from a primary table if matching records exist in a related table
- A primary key value cannot be changed in the primary table if that record has related records
- A value cannot be entered in the foreign key field of the related table that doesn't exist in the primary key of the primary table
- A null value can be entered in the foreign key, specifying that the records are unrelated

Referential Integrity Constraints

E.g., CREATE TABLE department(Dept_id int NOT NULL,Dept_Name varchar(255),PRIMARY KEY (Dept_id));

Output

Table created.

DESC department;

TABLE DEPARTMENT

Column	Null?	Type
DEPT_ID	NOT NULL	NUMBER
DEPT_NAME	-	VARCHAR2(255)

Download CSV

2 rows selected.

nt VALUES('1','CSE');

1 row(s) inserted.

Referential Integrity Constraints

INSERT INTO department VALUES('','ECE');

```
ORA-01400: cannot insert NULL into ("SQL_LHHGHTQALMCMCEVKLWKDDKTZZ"."DEPARTMENT"."DEPT_ID") ORA-06512: at "SYS.DBMS_SQL", line 1721
```

INSERT INTO department VALUES('2','ECE');

```
1 row(s) inserted.
```

SELECT * FROM department;

DEPT_ID	DEPT_NAME
1	CSE
2	ECE

Download CSV

2 rows selected.

LE employee(ID int NOT NULL,Name varchar(255),Dept_id int,PRIMARY KEY(ID),FOREIGN KEY(Dept_id) REFERENCES department(Dept_id));

Referential Integrity Constraints

```
Table created.
```

DESC employee;

TABLE EMPLOYEE

Column	Null?	Type
ID	NOT NULL	NUMBER
NAME	-	VARCHAR2(255)
DEPT_ID	-	NUMBER

Download CSV

3 rows selected.

INSERT INTO employee VALUES('101','Sindhu','1');

```
1 row(s) inserted.
```

INSERT INTO employee VALUES('106','Sandhya','2');

```
1 row(s) inserted.
```

employee VALUES('103','Sindhu','5');

```
ORA-02291: integrity constraint (SQL_LHHGHTQALMCMCEVKLWKDDKTZZ.SYS_C0048235749) violated - parent key not found ORA-06512: at "SYS.DBMS_SQL", line 1721
```

Referential Integrity Constraints

INSERT INTO employee VALUES('101','Ranjith','1');

```
ORA-00001: unique constraint (SQL_LHHGHTQALMCMCEVKLWKDDKTZZ.SYS_C0048235748) violated ORA-06512: at "SYS.DBMS_SQL", line 1721
```

SELECT * FROM employee;

ID	NAME	DEPT_ID
101	Sindhu	1
106	Sandhya	2

Download CSV
2 rows selected.

DROP TABLE department;

```
ORA-02449: unique/primary keys in table referenced by foreign keys
```

DROP TABLE employee;

```
Table dropped.
```

DROP TABLE department;

```
Table dropped.
```

Key Constraints

- Keys are the entity set that is used to identify an entity within its entity set uniquely
- An entity set can have multiple keys, out of which one key will be the primary key
- A primary key can contain a unique and null value in the relational table
- No two rows can have identical values with respect to key attribute

S.No.	Emp_id	Add
1	X1	KNP
2	X2	JHS
3	X3	LUK
3	X4	MUM

Duplication of data not allowed

STUDENT_ID	NAME	SEMESTER	AGE
101	Manish	1st	18
102	Rohit	3rd	19
103	Badal	5th	20
102	Amit	7th	21

Not allowed. Because all rows must be **unique**

Key Constraints

E.g., CREATE TABLE customers(ID int NOT NULL,Name varchar(20) NOT NULL,Age INT NOT NULL UNIQUE,PRIMARY KEY (ID));

Output

```
Table created.
```

INSERT INTO customers VALUES('1','Sindhu','26');

```
1 row(s) inserted.
```

INSERT INTO customers VALUES('2','Sandhya','26');

```
ORA-00001: unique constraint (SQL_LHHGHTQALMCMCEVKLWKDDKTZZ.SYS_C0048236289) violated ORA-06512: at "SYS.DBMS_SQL", line 1721
```

```
1 row(s) inserted.
```

ers;

TABLE CUSTOMERS

Column	Null?	Type
ID	NOT NULL	NUMBER
NAME	NOT NULL	VARCHAR2(20)
AGE	NOT NULL	NUMBER

Download CSV

3 rows selected.

Keys and Integrity Constraints

Summary

- Keys in DBMS
- Integrity Constraints
- Domain Constraints
- Entity Integrity Constraints
- Referential Integrity Constraints
- Key Constraints

Keys and Integrity Constraints

Try Yourself

- Create a Voters database with the following constraints

✓ Age >=18

✓ ID should be unique

✓ Phone number as primary key

✓ Aadhar number as foreign key

✓ Name should not be null

MODULE II CONSTRAINTS AND SQL COMMANDS